Pass Effectiveness Heatmap — Mean Gate Reduction (%)
per circuit family per individual pass
CNOT Chain
QFT
Oracle (BV)
Random Clifford
GHZ
Circuit Family
100.0
0.0
0.0
100.0
0.0
0.0
0.0
0.0
32.9
0.0
32.9
0.0
25.3
0.0
23.8
0.0
0.0
0.0
0.0
0.0
Commutative Cancellation
Optimize1q Gates
Commutation (Phase 2)
Greedy (Phase 1)
Transpiler Pass
Mean Gate Reduction (%)
100
80
60
40
20
0
Key finding: CommutativeCancellation matches our Phase 2 on Oracle/RandomClifford but both miss QFT/GHZ.
Neither isolated pass explains the full-pipeline advantage (template matching + routing + phase polynomial).